Optimal Routing for Quantum Networks

Implementation of Marcello Caleffi's Routing Algorithm for Quantum Network Architectures

Master in Quantum Computing - University POLITEHNICA of Bucharest

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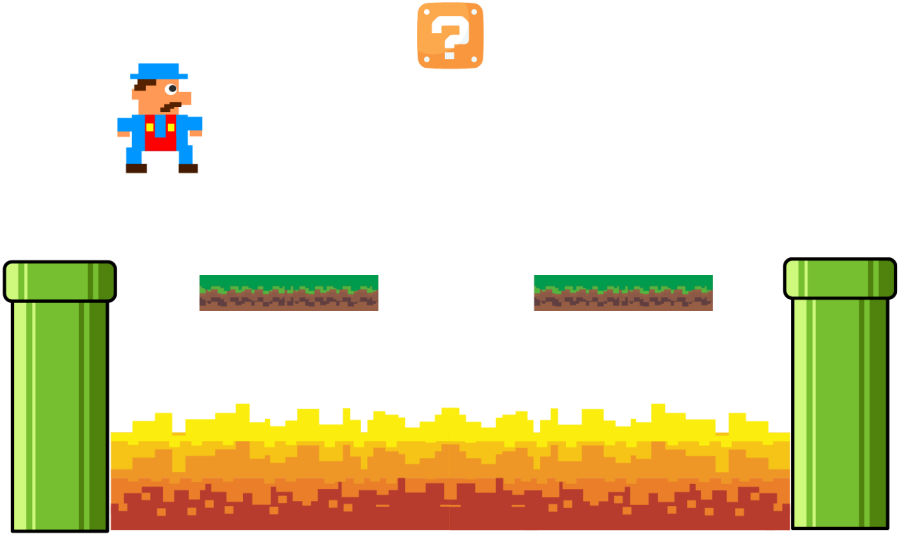
1. Problem Statement: Imperfect Quantum Networks

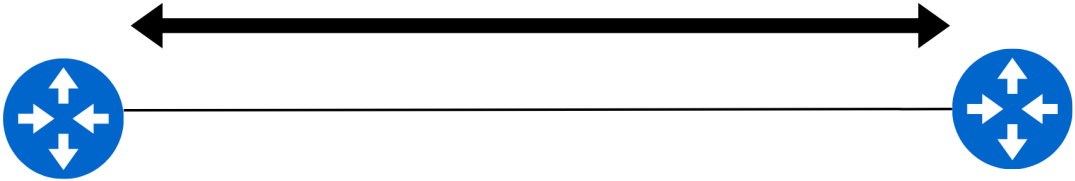
What is the issue with Quantum Networks?

Decoherence: The entanglement rate decays exponentially with link distance, while quantum memories can only preserve their stored qubits for a finite coherence time [Caleffi, 2017].

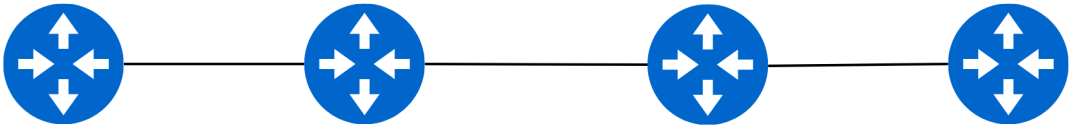
- We need quantum networks to connect quantum processors through remote entanglement distribution [Kimble, 2008]
- Is there a way to build an algorithm that is entanglement-aware and accounts for all physical conditions?

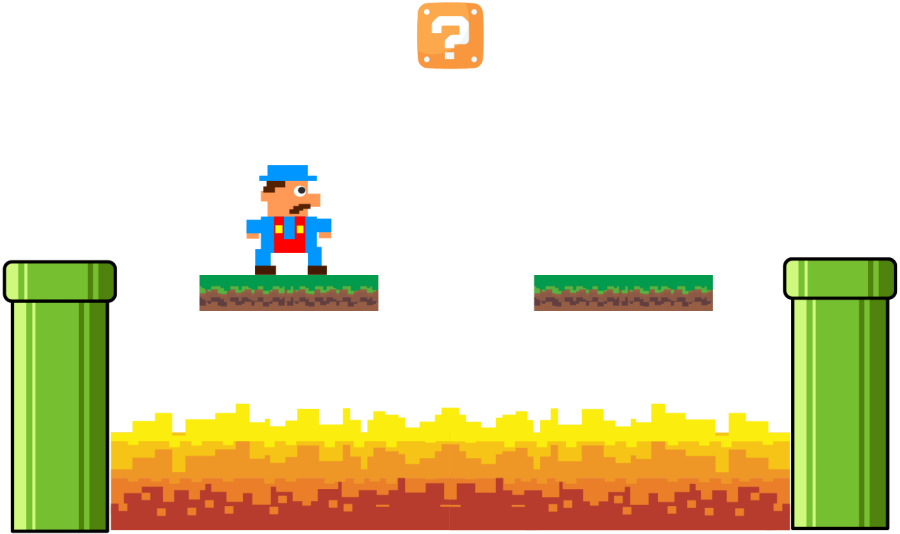


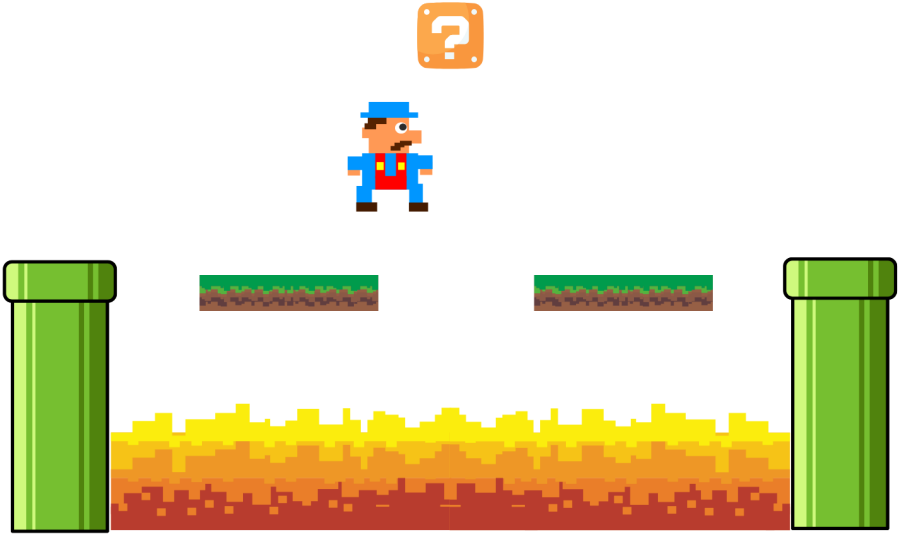
Long Links?!

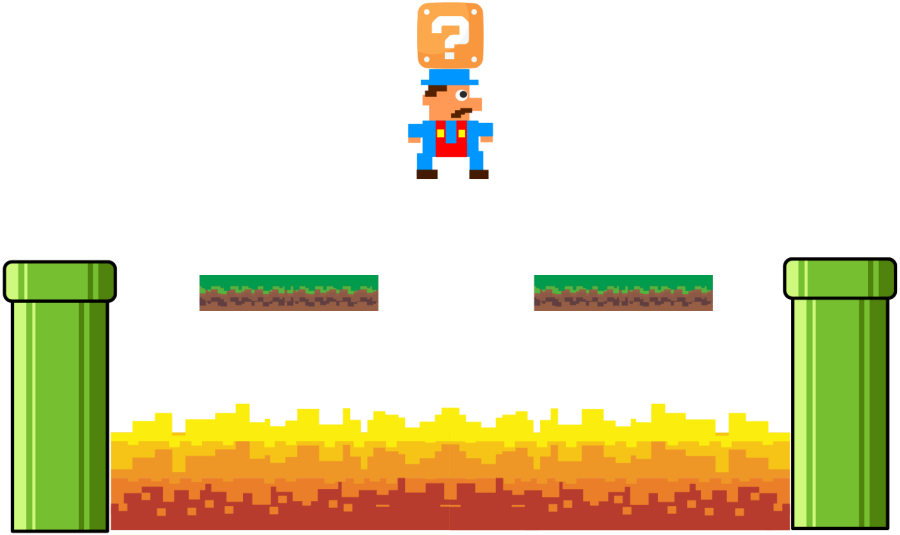


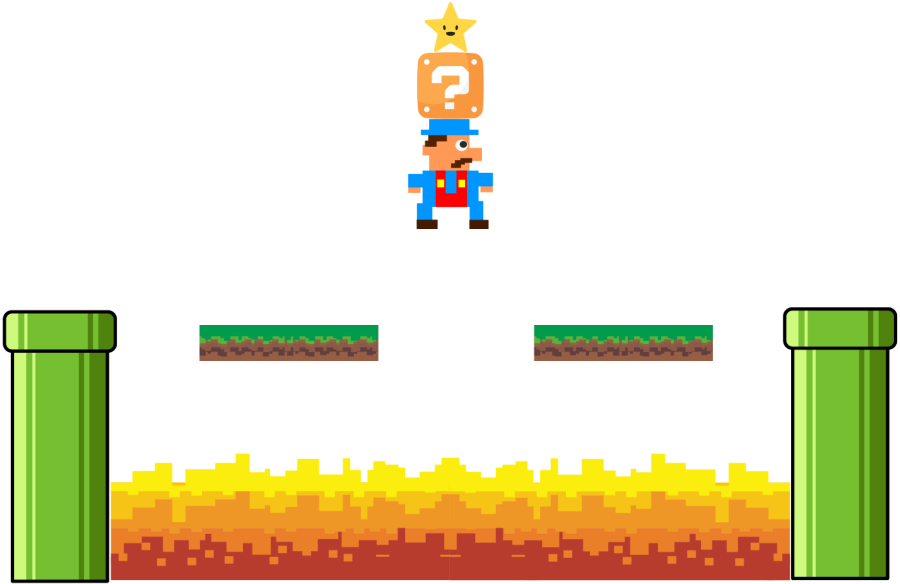
Shorter Links











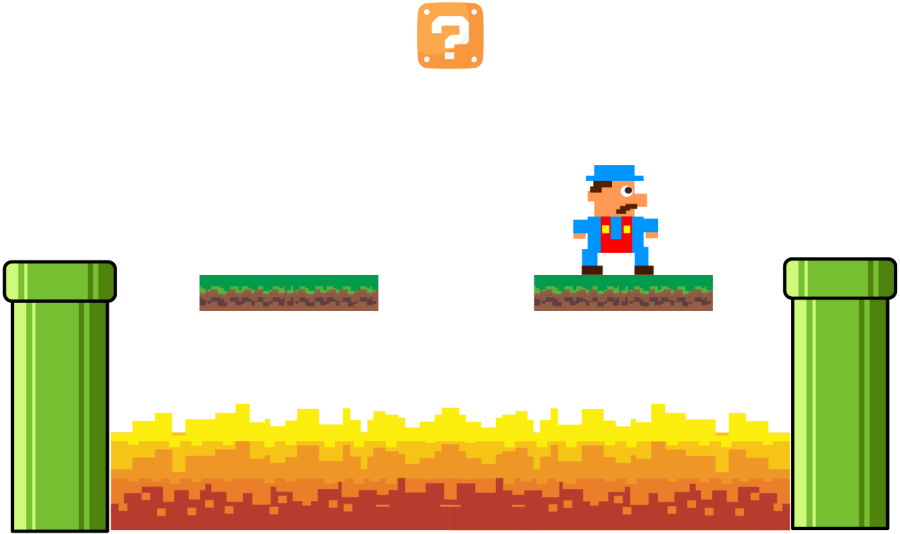
2. Quantum Repeaters

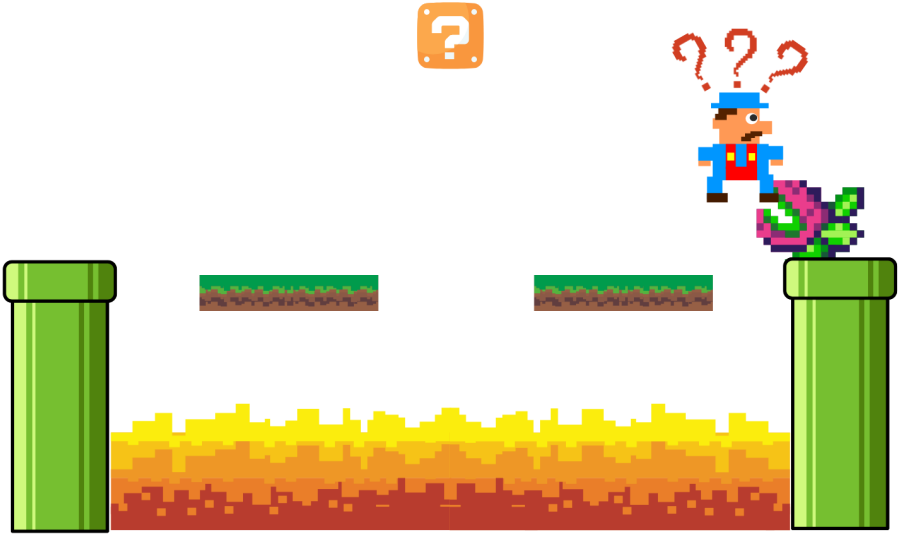
Entanglement Swapping

A Bell-state measurement performed at an intermediate node that fuses two adjacent entangled pairs into a single end-to-end entanglement between the two previously uncorrelated outer nodes (extending entanglement across non-adjacent repeaters) [Żukowski et al., 1993].

Entanglement Polarization

A distillation procedure that consumes several low-fidelity (noisy) entangled pairs to produce a smaller number of higher-fidelity pairs, counteracting channel noise and decoherence [Deutsch et al., 1996].





3. Physical Parameters Used in the Model

Paper symbol	Code name	Description
L_0	`l0`	Optical fiber attenuation length
p^{ht}	`p_ht`	Atom–photon entanglement probability
τ^p	`tau_p`	Atom pulse duration
τ^d	`tau_d`	Atom cooling cycle duration
ν^h/τ^h	`nu_h` / `tau_h`	Herald detector efficiency / duration
ν^t/τ^t	`nu_t` / `tau_t`	Telecom detector efficiency / duration
ν^o/τ^o	`nu_o` / `tau_o`	Optical BSM efficiency / duration
ν^a/τ^a	`nu_a` / `tau_a`	Atomic BSM efficiency / duration
T^{CH}	`t_ch`	Quantum memory coherence time
c_f	`cf`	Speed of light in fiber

4. Auxiliary Functions

Theorem: End-to-end Entanglement Rate [Caleffi, 2017]

The expected entanglement rate $\xi_{r_{i,j}}(T^{ch})$ between nodes v_i and v_j through route $r_{i,j}$ is given below, with T^{ch} the quantum memory coherence time and $\tau_{r_{\sigma_l,\sigma_m}}$ recursively defined as in eq. (12).

$$\xi_{r_{i,j}}(T^{ch}) = \begin{cases} 0, & \text{if } T^{ch} < \tau_{r_{i,j}} - \min\limits_{l=1,\,n-1} \left\{ T^{s}_{\sigma_l,\sigma_{l+1}} - \tau_{\sigma_l,\sigma_{l+1}} \right\} \\ \dfrac{1}{T_{r_{i,j}}}, & \text{otherwise} \end{cases}$$

4. Auxiliary Functions (1)

Minimum Required Coherence Time [Caleffi, 2017]

$\tau_{r_{\sigma_l,\sigma_m}}$ denotes the average duration of the successful (last) round of link-entanglement operations required to generate a remote entanglement between two non-adjacent nodes v_i and v_j through route $r_{i,j}$ — i.e., the minimum time the quantum memories must preserve coherence for the end-to-end entanglement to be usable.

$$\tau_{r_{\sigma_l,\sigma_m}} = \begin{cases} \max\left\{\tau_{r_{\sigma_l,\sigma_k}},\ \tau_{r_{\sigma_k,\sigma_m}}\right\} + \tau^a + \max\left\{T^c_{r_{\sigma_l,\sigma_k}},\ T^c_{r_{\sigma_k,\sigma_m}}\right\}, & k = \left\lceil \frac{m+l}{2} \right\rceil,\ \text{if } m > l+1 \\ T^s_{\sigma_l,\sigma_{l+1}}, & \text{otherwise} \end{cases}$$

5. Expected Entanglement Rate

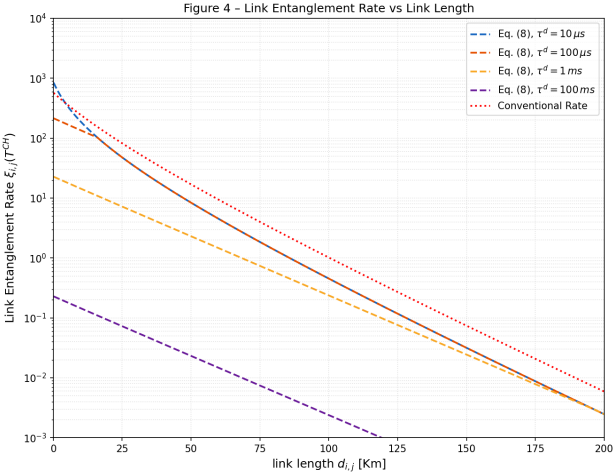
Algorithm 1 — Expected Entanglement Rate

function $\Xi(r_{i,j}, D)$
  for $e_{l,m} \in r_{i,j}$:
    compute $p_{l,m}$, $T_{l,m}$, $\tau_{l,m}$
  $n \leftarrow \text{numLinks}(r_{i,j})$
  if $n = 1$:
    if $\tau_{l,m} \leq T^{CH}$: $\xi_{r_{i,j}} \leftarrow 1/T_{l,m}$
  else:
    $k \leftarrow \lceil (n+1)/2 \rceil$
    $T_{r_{i,k}} \leftarrow \text{recT}(r_{i,k}, T, T^c)$; $T_{r_{k,j}} \leftarrow \text{recT}(r_{k,j}, T, T^c)$
    $\tilde{T} \leftarrow \max\{T_{r_{i,k}}, T_{r_{k,j}}\}$
    $\tilde{T}^c \leftarrow \max\{T^c_{r_{i,k}}, T^c_{r_{k,j}}\}$
    $T_{r_{i,j}} \leftarrow (\tilde{T} + \tau^a + \tilde{T}^c)/v^a$
    $\tau_{r_{i,k}} \leftarrow \text{recTau}(r_{i,k}, T^s, T^c)$; $\tau_{r_{k,j}} \leftarrow \text{recTau}(r_{k,j}, T^s, T^c)$
    $\tilde{\tau} \leftarrow \max\{\tau_{r_{i,k}}, \tau_{r_{k,j}}\}$
    $\tau_{r_{i,j}} \leftarrow \tilde{\tau} + \tau^a + \tilde{T}^c$
    if $\tau_{r_{i,j}} - \min\{T^s_{l,m} - \tau_{l,m}\} \leq T^{CH}$: $\xi_{r_{i,j}} \leftarrow 1/T_{r_{i,j}}$
    return $\xi_{r_{i,j}}$

Notation

L_0	fiber attenuation length
v_o, p	optical & detector efficiency factors
$p_{l,m}$	success prob. of entanglement on $e_{l,m}$
$T^c_{l,m}$	classical communication time on $e_{l,m}$
τ^t, τ^o	transmission / optical processing delays
τ^p	post-processing delay
τ^h	herald (BSM) delay
τ^d	decoherence-related delay
τ^a, v^a	swap delay and swap success prob.
$\tau_{l,m}$	coherence time used on link $e_{l,m}$
$T^s_{l,m}$	link time on success
$T^f_{l,m}$	link time on failure
$T_{l,m}$	expected entanglement time on $e_{l,m}$
$T_{r_{i,j}}$	expected entanglement time on route
$\tau_{r_{i,j}}$	total coherence time required by route
T^{CH}	qubit coherence (memory) time
$\xi_{r_{i,j}}$	expected entanglement rate of $r_{i,j}$

Figure 4 — Link Entanglement Rate vs Length

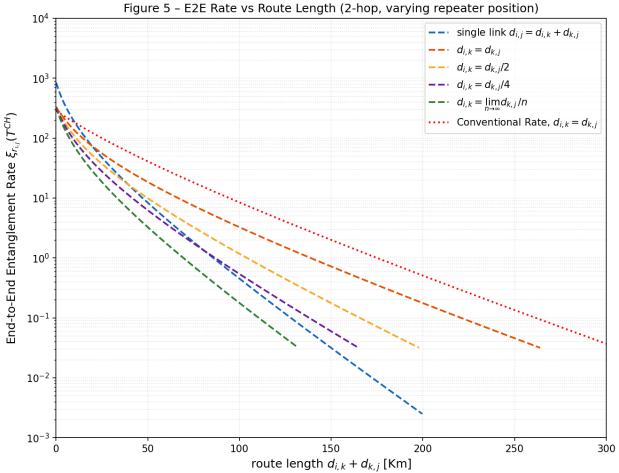


Physical parameters

Parameter	Value
τ^d (swept)	10 µs — 100 ms
T^{CH}	10 ms
τ_p	5.9 µs
τ_h	20 µs
τ_t	10 µs
τ_o	10 µs
τ_a	10 µs
p_{ht}	0.53
ν_h, ν_t	0.80
ν_o, ν_a	0.39
l_0	22 km
c_f	2×10^8 m/s

Red = swept across curves.

Figure 5 — End-to-End Rate (2-hop, varying repeater)

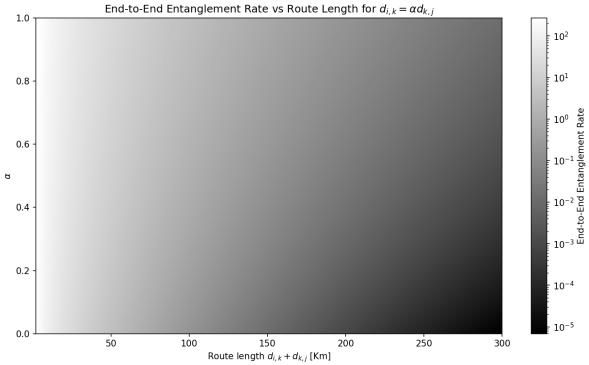


Physical parameters

Parameter	Value
τ^d	10 µs
T^{CH}	1.02 ms
τ_p	5.9 µs
τ_h	20 µs
τ_t	10 µs
τ_o	10 µs
τ_a	10 µs
p_{ht}	0.53
ν_h, ν_t	0.80
ν_o, ν_a	0.39
l_0	22 km
c_f	2×10^8 m/s

Red = overridden vs. defaults.

Figure 5 (heatmap) — Rate vs (D, α)

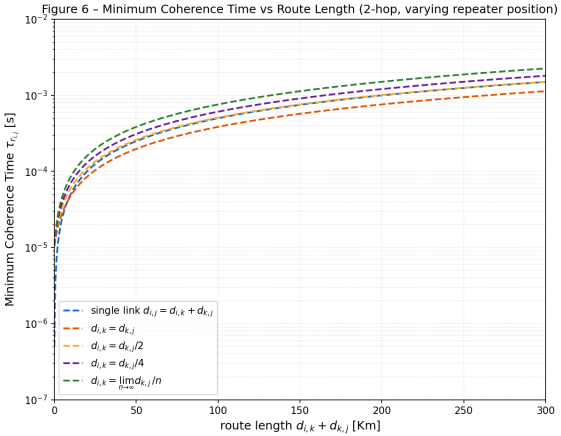


Physical parameters

Parameter	Value
τ^d	10 µs
T^{CH}	10 ms
τ_p	5.9 µs
τ_h	20 µs
τ_t	10 µs
τ_o	10 µs
τ_a	10 µs
p_{ht}	0.53
ν_h, ν_t	0.80
ν_o, ν_a	0.39
l_0	22 km
c_f	2×10^8 m/s

$\alpha = d_{i,k}/d_{k,j}$, D total route length.

Figure 6 — Minimum Coherence Time vs Route Length



Physical parameters

Parameter	Value
τ^d	10 µs
T^{CH}	1.02 ms
τ_t	0
τ_o	0
τ_p	5.9 µs
τ_h	20 µs
τ_a	10 µs
p_{ht}	0.53
ν_h, ν_t	0.80
ν_o, ν_a	0.39
l_0	22 km
c_f	2×10^8 m/s

Red = overridden. Switching/optical times zeroed for cleanest threshold curve.

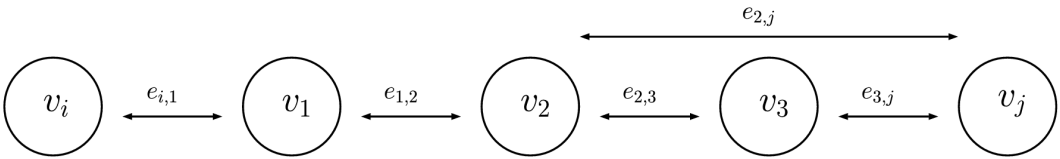
Figure 6 (heatmap) — Min Coherence Time vs (D, α)

Physical parameters

Parameter	Value
τ^d	10 µs
T^{CH}	1.02 ms
τ_t	0
τ_o	0
τ_p	5.9 µs
τ_h	20 µs
τ_a	10 µs
p_{ht}	0.53
ν_h, ν_t	0.80
ν_o, ν_a	0.39
l_0	22 km
c_f	2×10^8 m/s

$\alpha = d_{i,k}/d_{k,j}$, D total route length.

Use Case: Network Topology



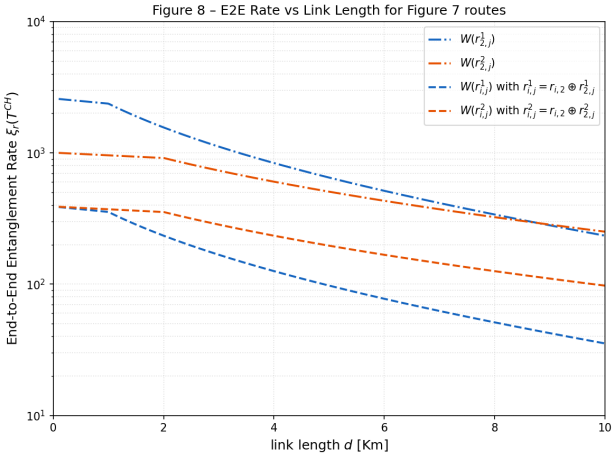
Network topology from [Caleffi, 2017]. There exist two simple routes from v_i to v_j:

i) $r^1_{i,j} = (e_{i,1},\ e_{1,2},\ e_{2,j})$,

ii) $r^2_{i,j} = (e_{i,1},\ e_{1,2},\ e_{2,3},\ e_{3,j})$,

resulting from the concatenation of the sub-route $r_{i,2} = (e_{i,1},\ e_{1,2})$ with the two routes $r^1_{2,j} = (e_{2,j})$ and $r^2_{2,j} = (e_{2,3},\ e_{3,j})$. The length of each link is d, with the exception of link $e_{2,j}$ which has length $2d$.

Figure 8 — E2E Rate for the Four Routes (Fig. 7)



Physical parameters

Parameter	Value
τ^d	10 μs
T^{CH}	10 ms
τ_t	5 μs
τ_o	5 μs
τ_p	5.9 μs
τ_h	20 μs
τ_a	10 μs
p_{ht}	0.53
ν_h, ν_t	0.80
ν_o, ν_a	0.39
l_0	22 km
c_f	2×10^8 m/s

Topology: v_i–v_1–v_2–v_3–v_j + shortcut $v_2 \to v_j$ of length $2d$.

6. Optimal Routing vs. Dijkstra

- **Algorithm 3** enumerates path combinations to globally maximize ξ.
- **Dijkstra** (with weight $1/\xi_{\text{link}}$) is a sub-optimal baseline since ξ is not isotonic.

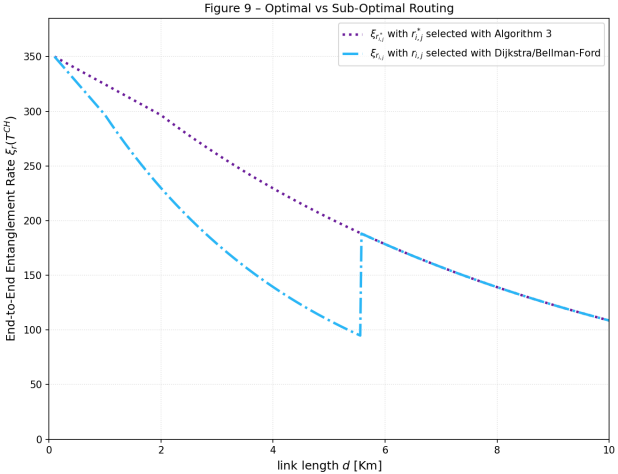
Algorithm 3 — Caleffi (2017) - $O(|V|^3 \cdot |R|^2)$

```
function optimalPath(V, E, D)
  for v_k ∈ V:
    for v_i ∈ V:
      for v_j ∈ V:
        for p_1 ∈ R(i, k), p_2 ∈ R(k, j):
          if ¬(V(p_1) ∩ V(p_2) \ {k}):
            r ← p_1 ⊕ p_2
            R(i, j).append(r)
            if Ξ(r, D) > w_{i,j}:
              r*_{i,j} ← r
              w_{i,j} ← Ξ(r, D)
  return {r*_{i,j}, w_{i,j}}_{v_i, v_j ∈ V}
```

Dijkstra Routing — $O((|V| + |E|) \log |V|)$

```
function dijkstraPath(V, E, s, t)
  for e_{i,j} ∈ E:  w(e_{i,j}) ← 1/ξ_{i,j}
  dist[v] ← ∞, ∀v ∈ V
  dist[s] ← 0;  prev[v] ← ⊥
  Q ← min-heap with (0, s)
  while Q ≠ ∅:
    (d, u) ← Q.pop()
    if u = t: break
    if d > dist[u]: continue
    for v ∈ adj(u):
      alt ← d + w(u, v)
      if alt < dist[v]:
        dist[v] ← alt;  prev[v] ← u
        Q.push((alt, v))
  return reconstructPath(prev, s, t)
```

Figure 9 — Optimal vs Sub-Optimal (Dijkstra) Routing



Physical parameters

Parameter	Value
τ^d	100 µs
T^{CH}	0.5 ms
τ_a	15 µs
l_0	8.15 km
ν_a	0.83
ν_o	1.0
τ_p	5.9 µs
τ_h, τ_t, τ_o	20/10/10 µs
p_{ht}	0.53
ν_h, ν_t	0.80
c_f	2×10^8 m/s

Red = overridden. l_0 tuned so $d \approx l_0 \ln 2 = 5.65$ km is the inflection.

7. Future Work

Validation & Benchmarking

- Evaluate the routing algorithm on a broader set of network topologies
- Stress-test under adversarial conditions: high link-loss rates, short coherence times, unbalanced path lengths

Integration with Quantum Network Simulators

- Validate against established frameworks: NetSquid[a], SeQUeNCe[b], SimulaQron[c]

[a] https://netsquid.org/
[b] https://github.com/sequence-toolbox/SeQUeNCe
[c] https://github.com/SoftwareQuTech/SimulaQron

7. Future Work (1)

Performance & Scalability

- Vectorize the entanglement-rate computation with NumPy[a]
- Port algorithm to Rust[b] or C++

[a] `https://numpy.org/`
[b] `https://www.rust-lang.org/`

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